

Supplementary Information

Permitless Concealed Carry Adoption and Firearm Mortality in U.S. States, 1999–2024

Yucheng Wang and Byung Kim
Both authors contributed equally to this work.

June 2026

Supplementary Note 1. Analytic hierarchy

Tier	Items	Role in interpretation
Primary	Firearm suicide	Main mortality endpoint.
Secondary	Total suicide; non-firearm suicide; firearm homicide; total firearm deaths	Specificity checks.
Sensitivity	Covariate layers; age-adjusted rates; legal edge cases; leave-one-out; placebo timing; COVID, weighting, trend, and staggered-adoption checks	Stability and limits.
Exploratory	Rurality, gun-ownership, baseline-risk, and policy-feature heterogeneity	Hypothesis generation only.
Audit trail	Full output files and legal-coding file	Reproducibility and source transparency.

Supplementary Note 2. Legal audit and edge-case handling

Audit status	State count	Primary-model treatment
Source-verified current adopter	26	Included as clean within-panel adopters.
Verified non-adopter	21	Included as verified controls.
Ambiguous reviewed	1	Excluded from primary model; Arkansas recoded separately.
Partial	1	Excluded from primary model; Mississippi reintroduced separately.
Baseline permitless	1	Excluded because Vermont has no within-panel adoption event.

State	Audit category	Reason for separate handling
Arkansas	Ambiguous reviewed	The clean annual treatment date depends on whether the relevant adoption is coded around 2021 or at the 2023 statutory clarification. Both codings are tested outside the primary model.
Mississippi	Partial	The 2015 change applied to narrower forms of concealed carry than the broader 2016 holster expansion. Mississippi is therefore not part of the clean-adoption headline estimate.
Vermont	Baseline permitless	Vermont had no carry-permit requirement before the panel began, so it cannot contribute a within-state adoption contrast.

Policy-feature field	State count
Training requirement removed	21
No prior training requirement to remove	5
Carry-permit background-check screen removed	25
Carry-permit screen removed while purchase-permit screen retained	1
Permit-specific misdemeanor screen removed	12
Eligibility standard retained but no pre-carry check	7
Other misdemeanor-screen coding categories	7

Supplementary Note 3. Main and secondary mortality estimates

Outcome	Coef.	SE	p	N
Firearm suicide	1.391	0.222	< 0.001	1222
Non-firearm suicide	0.415	0.139	0.003	1222
Total suicide	1.805	0.294	< 0.001	1222
Firearm homicide	-0.234	0.338	0.489	1142
Total firearm deaths	1.271	0.482	0.008	1222

Supplementary Note 4. Firearm-suicide specification and timing summary

Table 1: **Firearm-suicide diagnostic spine.**

Diagnostic row	Estimate	Diag. p	Interpretation
Baseline unweighted fixed effects	1.391	< 0.001	Clean-adoption state-level association
Population-weighted fixed effects	0.919	< 0.001	Resident-weighted association
Pre-COVID only	1.298	< 0.001	Drops post-2019 years
COVID-excluded	1.356	< 0.001	Drops 2020–2021
External firearm-law covariates	1.096	< 0.001	Policy-environment sensitivity
Leave-one-adopter-out	1.265–1.481	–	No sign changes across adopter exclusions
Never-treated placebo timing	1.391 > 0.338	–	Observed estimate exceeds placebo timing distribution
State-specific linear trends	0.249	0.079	Central attenuation caveat
Cohort ATT windows	0.601/0.577/0.641	–	2-, 3-, and 5-year windows
Not-yet-treated dynamic ATT	0.387	–	Positive post-adoption mean
Event-time fixed effects	0.651	0.0000001	Positive post mean; diagnostic p is the minimum pre-adoption p

Estimates are deaths per 100,000 residents. TWFE rows include state and year fixed effects, unemployment, per-capita income, and state-clustered standard errors unless otherwise noted. Modern staggered-adoption rows are reported as diagnostics because event-time checks flag pre-adoption signal; the event-time p column reports the minimum pre-adoption p rather than a post-adoption effect test.

Supplementary Note 5. Age-adjusted outcome-rate sensitivity

Outcome	Coef.	SE	p	N
Firearm suicide, age-adjusted	1.407	0.228	< 0.001	1220
Total suicide, age-adjusted	1.914	0.327	< 0.001	1222
Total firearm deaths, age-adjusted	1.230	0.510	0.016	1222
Firearm homicide, age-adjusted	-0.230	0.426	0.590	1037

Supplementary Note 6. Covariate-layer sensitivity

Firearm-suicide specification	Coef.	<i>p</i>	N
Baseline fixed effects	1.391	< 0.001	1222
External firearm-law covariates	1.096	< 0.001	1222
Health-access covariates	0.707	< 0.001	752
Overdose covariates	0.492	0.060	282
Demographic-poverty covariates	0.695	< 0.001	940
Mental-health-provider access covariates	0.116	0.744	141
Full Phase 3B2 covariates	0.176	0.633	141

Supplementary Note 7. Change-score checks

Outcome	2-year diff.	3-year diff.	5-year diff.
Firearm suicide	0.658**	0.634***	0.699***
Non-firearm suicide	0.214	0.134	0.047
Total suicide	0.872**	0.767***	0.746**
Firearm homicide	-0.034	0.024	0.035
Total firearm deaths	0.537	0.617	0.718

Supplementary Note 8. Staggered-adoption diagnostics

Outcome	ATT w2	ATT w3	ATT w5	Not-yet post	Interpretation
Firearm suicide	0.601	0.577	0.641	0.387	Positive, smaller than TWFE.
Non-firearm suicide	0.270	0.239	0.241	0.077	Positive, smaller than TWFE.
Total suicide	0.871	0.816	0.882	0.464	Positive post-adoption diagnostic with pretrend flag.
Firearm homicide	-0.004	-0.107	-0.106	-0.157	No positive post-adoption diagnostic.
Total firearm deaths	0.382	0.435	0.493	0.200	Positive post-adoption diagnostic with pretrend flag.

Supplementary Note 9. Robustness, influence, and legal sensitivity

Outcome	Baseline	P05 specs	Leave-one min	Leave-one max	Flag
Firearm suicide	1.391	5	1.265	1.481	Stable positive, trend-sensitive.
Non-firearm suicide	0.415	5	0.371	0.472	Stable positive, trend-sensitive.
Total suicide	1.805	6	1.656	1.954	Stable positive.
Firearm homicide	-0.234	0	-0.399	-0.049	No stable positive signal.
Total firearm deaths	1.271	4	1.058	1.532	Positive, trend-sensitive.

Table 2: **Firearm-suicide leave-one-adopter-out estimates by excluded state.**

Excluded state	Coef.	<i>p</i>
Alabama	1.415	< 0.001
Alaska	1.405	< 0.001
Arizona	1.481	< 0.001
Florida	1.398	< 0.001
Georgia	1.413	< 0.001
Idaho	1.375	< 0.001
Indiana	1.393	< 0.001
Iowa	1.395	< 0.001
Kansas	1.414	< 0.001
Kentucky	1.451	< 0.001
Louisiana	1.411	< 0.001
Maine	1.399	< 0.001
Missouri	1.366	< 0.001
Montana	1.342	< 0.001
Nebraska	1.404	< 0.001
New Hampshire	1.391	< 0.001
North Dakota	1.358	< 0.001
Ohio	1.410	< 0.001
Oklahoma	1.383	< 0.001
South Carolina	1.406	< 0.001
South Dakota	1.403	< 0.001
Tennessee	1.416	< 0.001
Texas	1.415	< 0.001
Utah	1.424	< 0.001
West Virginia	1.420	< 0.001
Wyoming	1.265	< 0.001

Table 3: **Firearm-suicide never-treated placebo timing estimates.**

Placebo shift	Coef.	<i>p</i>
0	-0.026	0.850
1	-0.246	0.173
2	-0.062	0.788
3	0.081	0.721
4	0.385	0.060
5	0.291	0.047
6	0.054	0.675
7	0.050	0.767
8	-0.032	0.883
9	-0.224	0.375
10	-0.275	0.156

Supplementary Note 10. Heterogeneity and policy-feature checks

Outcome/dimension	Interaction	<i>p</i>
Firearm suicide, high rurality	0.988	< 0.001
Total suicide, high rurality	1.733	< 0.001
Non-firearm suicide, high rurality	0.745	< 0.001
Non-firearm suicide, high gun ownership	0.586	< 0.001

Supplementary Note 11. Firearm-homicide missingness

Table 4: **Firearm-homicide missingness audit.**

Group	Possible	Observed	Missing	Missing pct.
All state-years	1222	1142	80	6.5%
Adopter-state state-years	676	610	66	9.8%
Non-adopter-state state-years	546	532	14	2.6%
Q1 smallest	312	232	80	25.6%
Q2	312	312	0	0.0%
Q3	286	286	0	0.0%
Q4 largest	312	312	0	0.0%

Missing firearm-homicide rates are concentrated in the smallest-population state-years. No state-year in population quartiles 2–4 is missing the firearm-homicide rate.

State	Missing	Possible	Missing pct.
New Hampshire	16	26	61.5%
North Dakota	15	26	57.7%
South Dakota	14	26	53.8%
Wyoming	14	26	53.8%
Hawaii	13	26	50.0%
Maine	6	26	23.1%
Delaware	1	26	3.8%
Montana	1	26	3.8%

Supplementary Note 12. Diagnostic source files

- `outputs/tables/did/twfe.did.main.results.csv`
- `outputs/tables/did/twfe.did.ageadj.sensitivity.summary.csv`
- `outputs/tables/did/twfe.did.firearm.law.control.summary.csv`
- `outputs/tables/did/twfe.did.nonfirearm.confounder.summary.csv`
- `outputs/tables/did/twfe.did.phase3b2.confounder.summary.csv`
- `outputs/tables/main/welch.change.score.results.csv`
- `outputs/tables/modern.did/modern.did.summary.csv`
- `outputs/tables/modern.did/event.time.never.control.estimates.csv`
- `outputs/tables/robustness/robustness.summary.csv`
- `outputs/tables/robustness/leave.one.state.out.csv`
- `outputs/tables/robustness/placebo.never.treated.csv`
- `outputs/tables/robustness/arkansas.treatment.sensitivity.summary.csv`
- `outputs/tables/heterogeneity/heterogeneity.did.results.csv`
- `outputs/tables/mechanism/mechanism.heterogeneity.results.csv`
- `data/policy/permitless.carry.legal.audit.csv`